

Vocal Acoustic Biomarker Analysis for Early Parkinson's Disease Detection: A Machine Learning Approach

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Introduction

Parkinson's Disease (PD) is a progressive neurodegenerative disorder affecting movement, speech, and quality of life. Current diagnosis relies on clinical evaluation of motor symptoms, which often do not appear until significant neurological damage has already occurred, limiting opportunities for early intervention. This motivates the search for non-invasive, accessible methods for earlier detection.

One promising approach involves vocal acoustic biomarkers. Parkinson's frequently disrupts speech production, causing measurable changes in pitch, jitter, shimmer, and voice intensity that can precede severe motor symptoms. Voice recordings can be collected non-invasively using smartphones or microphones, making them a scalable and low-cost source of diagnostic information. Literature suggests these biomarkers reliably reflect disease-related changes, supporting their use as a basis for machine learning-based classification.

Problem

This project addresses the development of a machine learning model to classify Parkinson's Disease from vocal acoustic measurements, a supervised binary classification task. Input features consist of extracted acoustic variables from voice recordings, with the target label indicating disease status.

This work sits within the broader challenge of early disease detection and digital health monitoring. Voice-based Parkinson's detection is a relevant biomedical application because it combines signal processing and machine learning to provide an objective, scalable, and non-invasive screening tool that could improve accessibility and earliness of diagnosis.

Method

The analysis pipeline was implemented in a Google Colab notebook using Python, with cells run for preprocessing, PCA/correlation analysis, cross-validation, model training, and evaluation. The Parkinson's disease voice dataset consisted of 195 voice recordings from 31 subjects, with each recording represented by 22 acoustic voice features and a binary disease status label (1 = Parkinson's disease, 0 = healthy control). Prior to model development, acoustic features were separated from the target labels, and subject identifiers were extracted from the

recording names. Because multiple recordings existed per subject, subject IDs were used during cross-validation to prevent data leakage between training and testing sets.

Feature standardization was performed using z-score normalization with Scikit-learn's StandardScaler. Within each cross-validation fold, the scaler was fit only on training data and then applied to test data to prevent information leakage.

Principal component analysis (PCA) and correlation analysis were performed on the standardized dataset to examine variance structure and identify redundancy among acoustic features. Based on these analyses, a subset of 10 features was selected for neural network training: MDVP(dB), MDVP(%), HNR, NHR, RPDE, DFA, spread1, spread2, D2, and PPE. Logistic regression was trained using all 22 acoustic features as a baseline comparison model.

Model evaluation used stratified grouped 5-fold cross-validation implemented with Scikit-learn's StratifiedGroupKFold. Grouping ensured that all recordings from the same subject remained within a single fold, while stratification maintained similar proportions of Parkinson's and healthy recordings across folds.

The primary predictive model was a feedforward neural network implemented in TensorFlow/Keras with two hidden layers (64 and 32 neurons), ReLU activations, dropout regularization, and a sigmoid output layer. The model was trained using the Adam optimizer and binary cross-entropy loss.

Logistic regression was implemented using Scikit-learn's LogisticRegression. Model performance was evaluated using accuracy, F1 score, receiver operating characteristic (ROC) curves, and area under the ROC curve (AUC). Unit and integration tests were included to verify correct implementation of cross-validation, model training, prediction, and performance evaluation.

Project files were organized into code, data, results, notebooks, tests, and documentation folders, with a README describing dependencies and instructions for running the workflow.

Results

Projection of the standardized acoustic voice features onto the first two principal components showed partial separation between Parkinson's disease and healthy recordings, with some overlap between groups (Figure 1). This pattern suggests that the voice features capture disease-related differences, but that classification requires additional model-based analysis beyond visualization of the first two principal components.

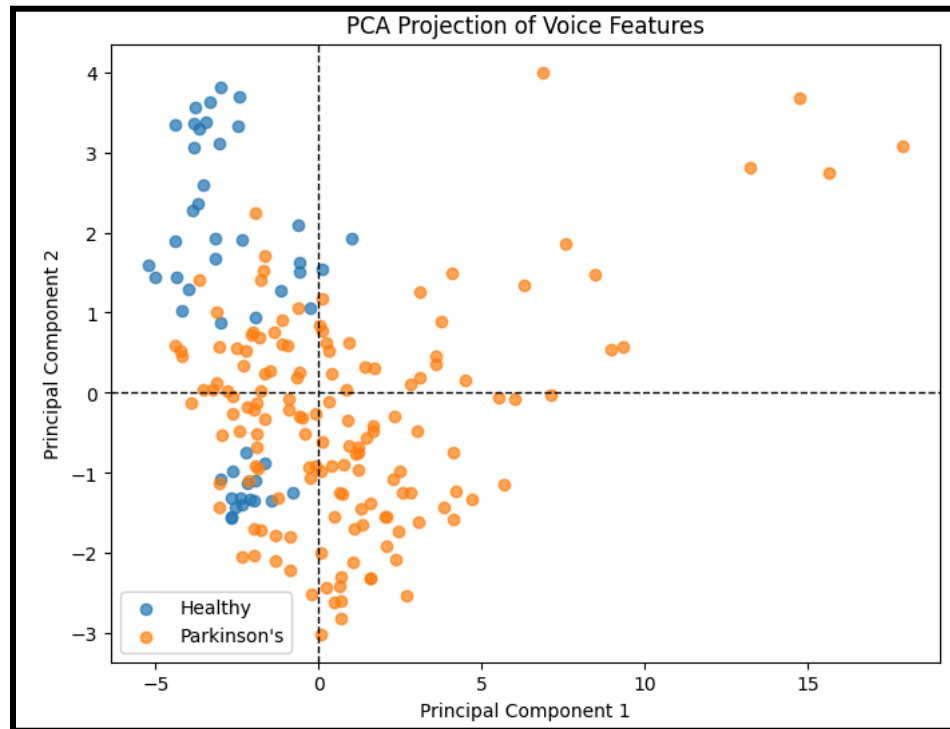


Figure 1. PCA projection of the Parkinson's disease voice dataset using the first two principal components. The plot shows partial separation between healthy and Parkinson's recordings, although overlap remains between groups.

Principal component analysis (PCA) was applied to the standardized feature matrix to examine variance structure. Six principal components captured approximately 90% of the total variance, while eight captured approximately 95% (Figure 2), indicating that much of the information in the original feature set can be represented by a smaller number of latent variables.

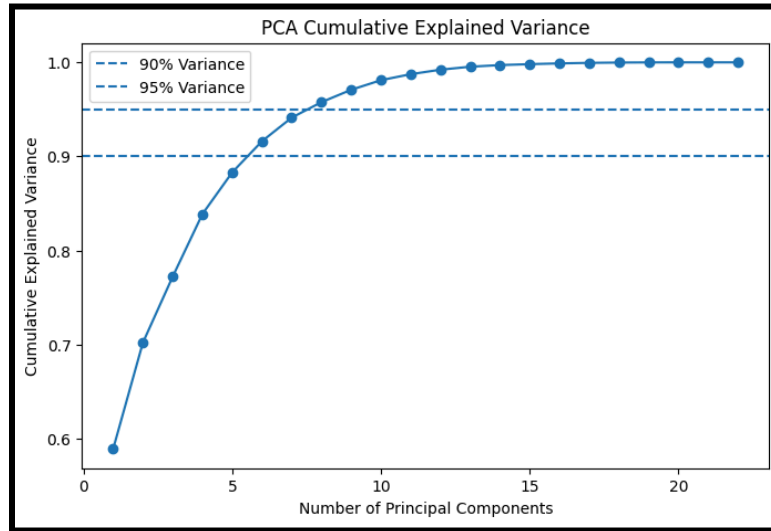


Figure 2. Cumulative explained variance captured by principal components derived from the 22 acoustic voice features.

Model performance was evaluated using stratified grouped 5-fold cross-validation to prevent subject-level data leakage. The neural network achieved a mean cross-validated accuracy of $78.6\% \pm 17.0\%$, a mean F1 score of 0.844 ± 0.146 , and an overall grouped cross-validated ROC-AUC of 0.856. The logistic regression model achieved a mean cross-validated accuracy of $80.3\% \pm 12.7\%$, a mean F1 score of 0.862 ± 0.096 , and an overall grouped cross-validated ROC-AUC of 0.738.

Model	Accuracy (%)	F1 Score	ROC-AUC
Neural Network	78.6 ± 17.0	0.844 ± 0.146	0.856
Logistic Regression	80.3 ± 12.7	0.862 ± 0.096	0.738

Table 1. Cross-validated model performance.

Fold-level neural network performance varied from an AUC of 0.572 to 0.981, reflecting differences among subject groups assigned to each test fold.

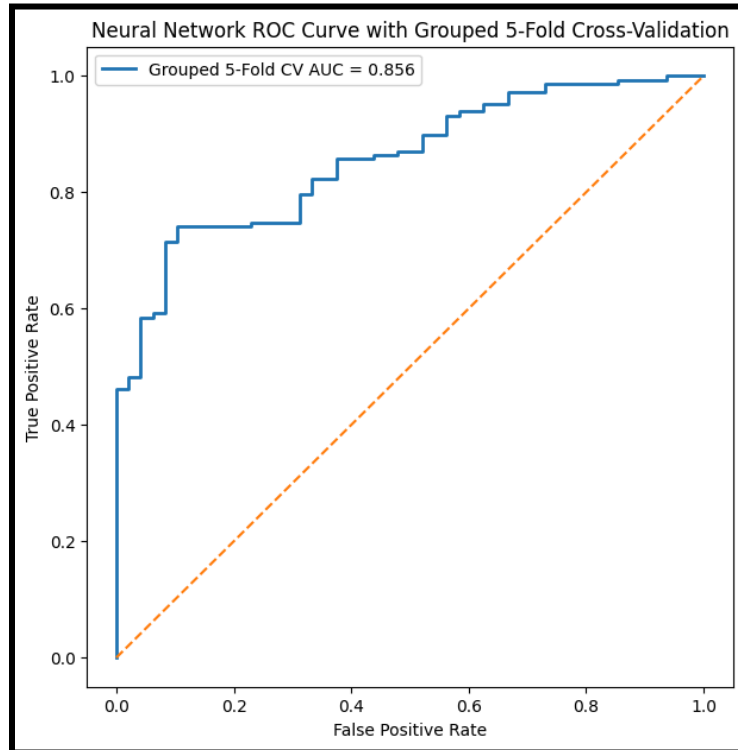


Figure 3. Neural network ROC curve generated using grouped 5-fold cross-validation.

ROC analysis showed stronger overall discrimination for the neural network than logistic regression, as reflected by the higher ROC-AUC (Figure 4).

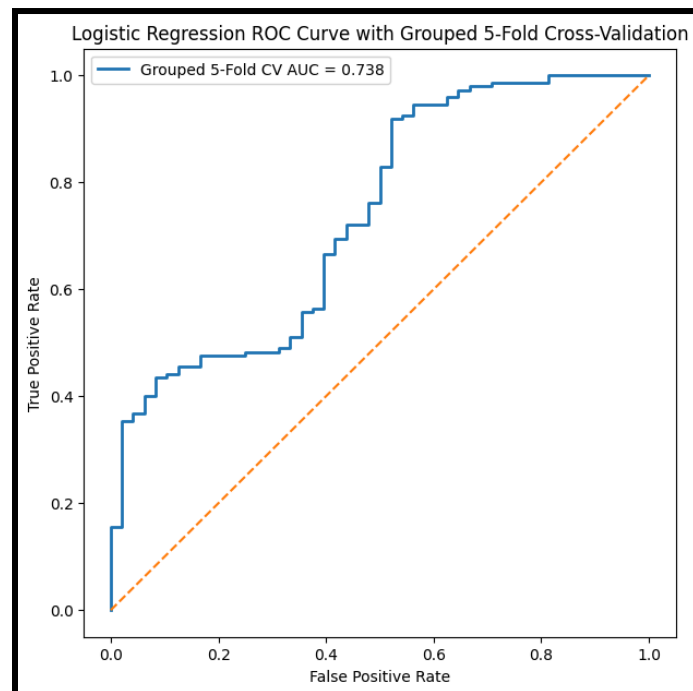


Figure 4. Logistic regression ROC curve generated using grouped 5-fold cross-validation.

Feature contribution analysis identified spread1, PPE, and MDVP as the highest-contributing predictors based on absolute logistic regression coefficient magnitude (Figure 5).

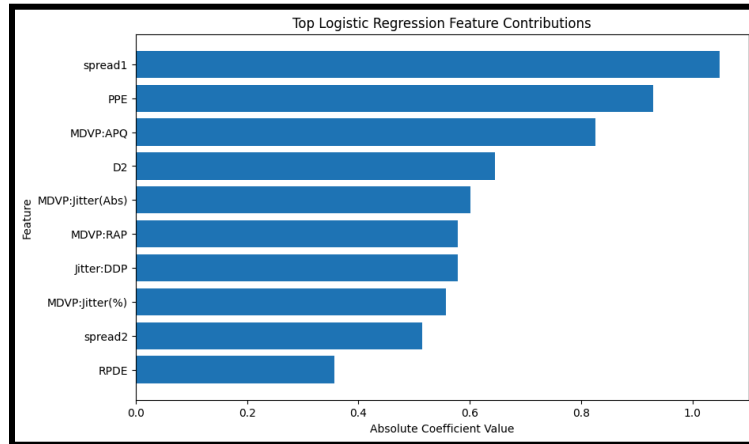


Figure 5. Top 10 acoustic voice features ranked by absolute logistic regression coefficient magnitude.

Together, these results demonstrate that acoustic vocal biomarkers contain meaningful predictive information for Parkinson's disease classification.

Discussion

This study demonstrates that acoustic voice measurements contain sufficient information to distinguish individuals with Parkinson's disease from healthy controls using machine learning methods. PCA revealed substantial redundancy among the 22 acoustic features, with a small number of principal components capturing most of the dataset variance. The partial separation observed in PCA space further suggested that disease-related information was present within the vocal measurements prior to model training.

Comparison of the neural network and logistic regression models highlighted the importance of nonlinear relationships among acoustic biomarkers. Although logistic regression achieved slightly higher mean accuracy and F1 score, the neural network produced a higher ROC-AUC, indicating stronger overall discrimination between Parkinson's and healthy recordings. This suggests that disease-related vocal patterns are not fully captured by a linear

decision boundary. Feature contribution analysis identified spread1, PPE, and MDVP as the most influential predictors in the logistic regression model. These features are associated with vocal instability and irregular speech production, both characteristic symptoms of Parkinson's disease.

Several limitations should be considered when interpreting these results. First, the dataset contained only 31 unique subjects, limiting the diversity of vocal characteristics available for training. Although multiple recordings were collected from each individual, the number of independent subjects remained small, reducing confidence in how well the models would generalize to broader patient populations. Second, the dataset only distinguished Parkinson's disease from healthy controls. Since vocal symptoms such as tremor, instability, and irregular speech production may also occur from non-parkinsonian sources, additional studies are needed to determine whether these models can distinguish Parkinson's-specific vocal patterns from similar symptoms caused by other disorders. Finally, the models relied on pre-extracted acoustic features rather than raw speech recordings, potentially excluding useful temporal and spectral information present in the original audio signal.

Conclusion

In this study, machine learning techniques were applied to acoustic voice recordings to classify individuals with Parkinson's disease and healthy controls. Principal component analysis demonstrated that the dataset contained substantial feature redundancy while preserving meaningful class structure. A neural network model trained on a selected subset of acoustic features achieved stronger discriminative performance than logistic regression, producing a higher cross-validated ROC-AUC despite comparable accuracy and F1 scores. The findings demonstrate that vocal biomarkers contain predictive information relevant to Parkinson's disease detection and that nonlinear machine learning models can effectively leverage these signals. While additional validation on larger and more diverse datasets is required, the results support the feasibility of non-invasive, voice-based screening approaches for Parkinson's disease. Such tools may ultimately contribute to more accessible monitoring and earlier identification of neurological impairment in both clinical and remote healthcare settings.

References

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Team member contributions

Vibha Doddipalle: Feature contributions visualizations

Ila Gowda: PCA Visualizations

Madeline Kuan: Neural Network

Lindsey Lee: Neural Network

Noah Trejo: Logistic Regression